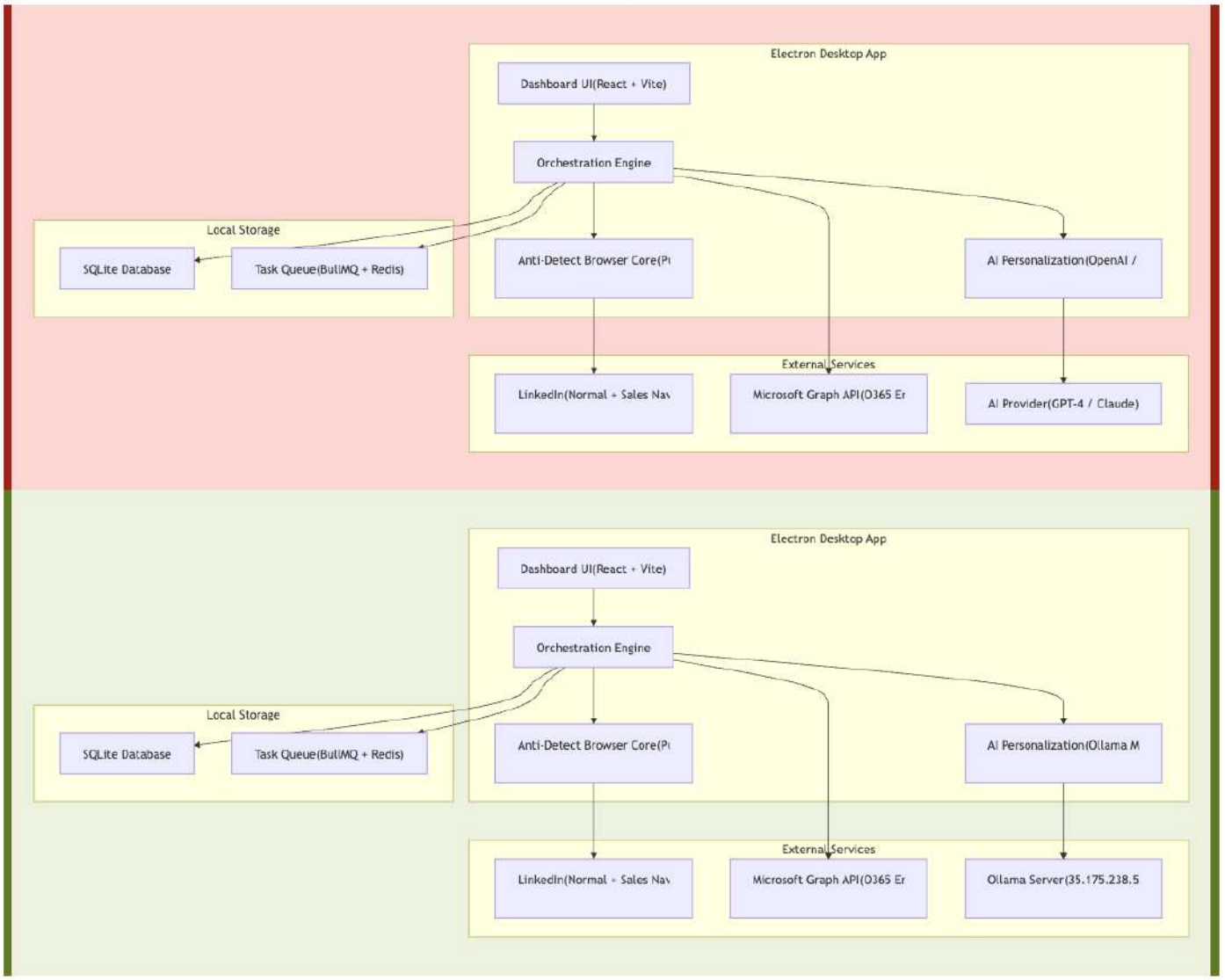
VirtuLinked AI — LinkedIn Automation Platform

A desktop-first LinkedIn automation tool inspired by LinkedHelper's safety-first approach, with deep Microsoft 365 integration for email, calendar, and AI-powered personalization.

1. High-Level Architecture



Why Desktop-First (Like LinkedHelper)?

Aspect	Desktop (Our Choice)	Cloud-Based
Detection Risk	✅ Low — runs from your IP, single session	❌ High — shared IPs, multiple sessions
Browser Fingerprint	✅ Unique, matches your machine	❌ Data center fingerprints flagged
Session Management	✅ Persistent cookies, natural login	❌ Cookie rotation risks
LinkedIn Compliance	✅ Mimics real user on real machine	❌ Clearly automated infrastructure

LinkedIn Compliance	✅ Mimics real user on real machine	❌ Clearly automated infrastructure
Data Privacy	✅ All data stays local	❌ Credentials on third-party servers

2. Core Modules Breakdown

Module A: Anti-Detect Browser Engine

The foundation that makes everything safe. This is what LinkedHelper does best.

How It Works:

- Electron app embeds **Puppeteer** with `puppeteer-extra-plugin-stealth`
- No code injection into LinkedIn pages — all interactions via DOM simulation
- In-page navigation only (no direct URL jumps — LinkedIn detects these)

Safety Features:

Feature	Implementation
Randomized Delays	Gaussian distribution between actions (2–8s for clicks, 50–150ms per keystroke)
Human-Like Typing	Variable speed, occasional "typos" corrected, natural pauses
Mouse Movement	Bézier curve movements to targets, random hovering
Working Hours	Configurable schedule (e.g., 9AM–6PM IST) with ±30min randomized start
Daily Limits	Smart limits with ±15% daily randomization
Session Management	Natural login/logout cycles, session persistence between runs
Scroll Behavior	Variable speed scrolling, pauses at content, natural reading patterns
Browser Fingerprint	Consistent fingerprint matching your real machine

Safe Daily Action Limits (Configurable):

Action	Default Limit	Range
Connection Requests	20–25/day	Max 100/week

Action	Default Limit	Range
Connection Requests	20–25/day	Max 100/week
Profile Views	50–80/day	Varies by account type
Messages	30–50/day	Respects LinkedIn caps
Post Engagements	20–30/day	Likes, comments
Content Posts	1–2/day	Scheduled

> IMPORTANT
> **Account Warm-Up Protocol:** New accounts start at 20% of limits and ramp up over 2–3 weeks. This prevents the #1 cause of LinkedIn bans — sudden activity spikes.

Module B: LinkedIn Interaction Layer

Two sub-modules for Normal Account + Sales Navigator:

B1. Profile Scraper & Analyzer

- Visit profiles naturally (scroll, pause, read sections)
- Extract: Name, headline, current role, company, experience, education, skills, recent posts, mutual connections, about section
- For Sales Navigator: additional filters, lead lists, account insights
- Store all data locally in SQLite

B2. Connection Request Manager

- **Smart Note Personalization:** AI reads profile data → generates personalized note (≤300 chars)
- **Conditional Logic:**
  - If note can be sent → send with personalized note
  - If note can't be sent (e.g., limit reached) → send without note + queue personalized email
- **Spintax Support:** Multiple message templates with variable rotation
- **Queue Management:** Respects daily limits, prioritizes by lead score

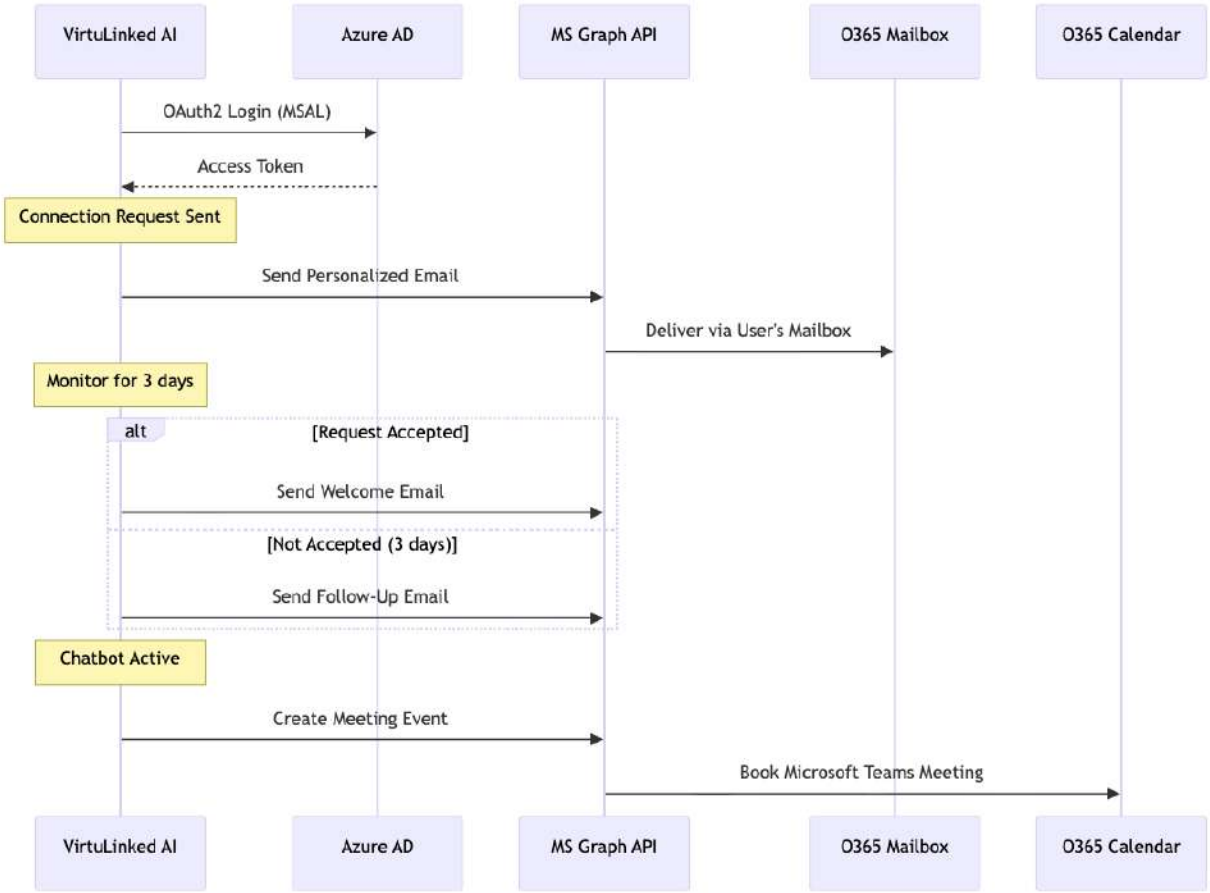
B3. LinkedIn Messaging (Post-Connection)

- Auto-reply chatbot for LinkedIn DMs
- Conversational AI that guides toward booking a meeting
- Natural response delays (2–15 minutes, configurable)
- Handoff detection: stops auto-reply when user manually intervenes

---

Module C: Microsoft 365 Integration (Graph API)

Module C: Microsoft 365 Integration (Graph API)



C1. O365 Email Integration

- **Azure App Registration** with `Mail.Send`, `Mail.ReadWrite`, `Calendars.ReadWrite` permissions
- **OAuth2 PKCE flow** for secure desktop auth (no client secrets stored)
- Send HTML emails from your actual mailbox (not spoofed)
- Track email opens/clicks (via tracking pixel + link wrapping)
- Template system with dynamic variables from profile data

C2. Calendar Meeting Booking

- Create Microsoft Teams meetings programmatically
- Generate booking links with available time slots
- Auto-insert meeting links in chatbot conversations
- Meeting confirmation and reminder emails

C3. Email Sequences

Trigger	Timing	Email Type
Connection request sent	Immediately	Personalized intro email tied to your services
	Within 1	

Connection request sent	Immediately	Personalized intro email tied to your services
Request accepted	Within 1 hour	Welcome + value proposition email
Request NOT accepted	After 3 days	Gentle follow-up email
Chatbot conversation active	Dynamic	Meeting booking prompt
Meeting booked	Immediately	Confirmation + agenda email

Module D: AI Personalization Engine

Purpose: Generate hyper-personalized content that doesn't look automated.

How It Works:

1. Profile data → AI prompt engineering → Personalized output
2. Uses OpenAI GPT-4 / Claude API (configurable)
3. Every message is unique — no two leads get the same text

1. Profile data → AI prompt engineering → Personalized output
2. Primary: **Ollama Mixtral 8x7B** on self-hosted server (`35.175.238.52`)
3. Fallback model: **Mistral 7B Instruct** (lighter, faster) on same server
4. Optional: Cloud AI (OpenAI/Anthropic) toggle in settings
5. Every message is unique — no two leads get the same text

Ollama Server Configuration:

Endpoint	URL	Purpose
Tags/Models	`http://35.175.238.52:11434/api/tags`	List available models
Generate	`http://35.175.238.52:8080/api/generate`	Text generation endpoint

Available Models:

Model	Size	Use Case
`mixtral:8x7b-instruct-v0.1-q3_K_M`	22.5 GB	Primary — connection notes, emails, chatbot, comments
`mistral:7b-instruct-v0.3-q4_K_M`	4.4 GB	Fast fallback — quick replies, sentiment analysis

Personalization Variables:

Variable	Source	Example
`{firstName}`	Profile	"Akash"
`{company}`	Profile	"Veda AI Lab"
`{role}`	Profile	"CTO"
`{recentPost}`	Activity	"Your post about AI in healthcare..."
`{mutualConnection}`	Network	"We both know Priya Sharma"
`{skillMatch}`	Skills	"Your expertise in ML alignment..."
`{industryInsight}`	AI Generated	Relevant industry observation

AI-Generated Content Types:

- Connection request notes (≤300 chars, human tone)
- Personalized emails (matched to your services)
- Follow-up sequences (context-aware)
- LinkedIn DM replies (conversational, goal-oriented)
- Post comments (genuine, insightful)

Module E: Content Scheduler & Engagement Automation

E1. Auto Content Scheduler

- **Post Types:** Text, image, video, carousel, polls, articles
- **Scheduling:** Calendar-based scheduling with optimal posting times
- **Content Queue:** Draft, review, schedule, publish pipeline
- **Best Time Detection:** Learns when your audience is most active
- **Multi-format:** Adapts content for different post types

E2. Engagement Enhancement ("Make It Viral")

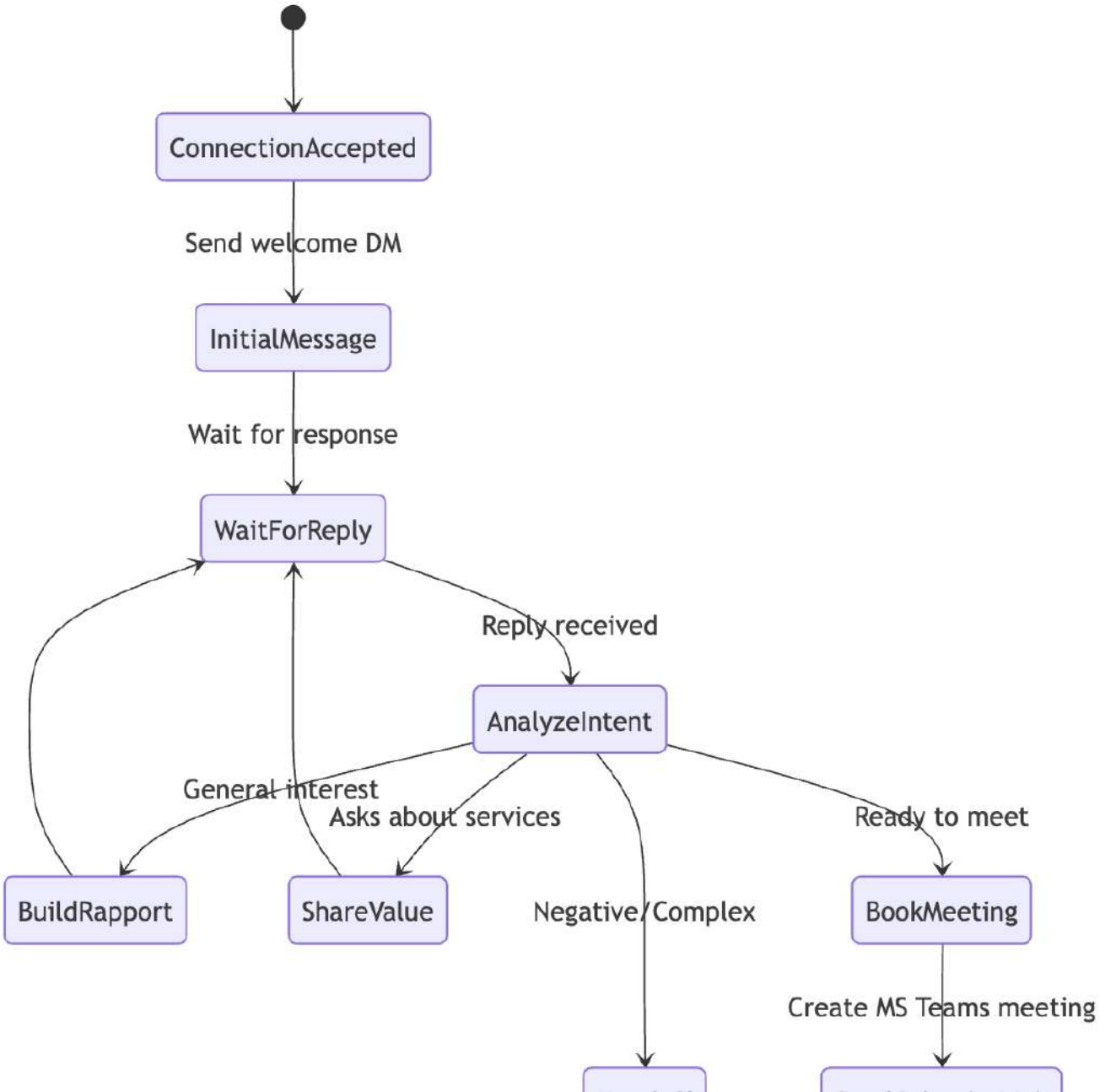
- **Auto-Like:** Smart engagement on target audience's posts
- **Auto-Comment:** AI-generated relevant comments (not generic "Great post!")
- **Comment Reply:** Auto-reply to comments on YOUR posts with AI
- **Engagement Pods:** Coordinate engagement timing for maximum reach
- **Hashtag Optimization:** AI-suggested hashtags based on content + trends
- **Repost Strategy:** Auto-share industry content with personalized commentary

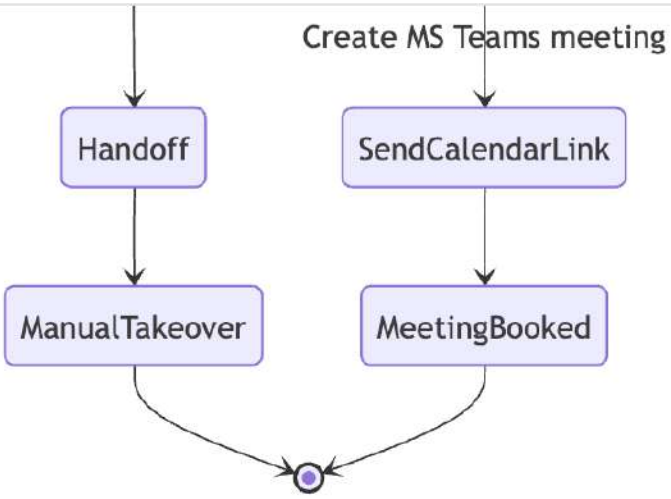
E3. Engagement Safety

E3. Engagement Safety

Feature	Implementation
Comment Quality	AI generates unique, contextual comments (min 2 sentences)
Engagement Spread	Actions distributed across the day, not burst
Content Variety	Mix of likes, comments, shares — not just one type
Target Relevance	Only engage with content in your industry/niche
Cool-Down Periods	Random breaks between engagement sessions

Module F: Auto-Reply Chatbot





Key Behaviors:

- Response delay: 2–15 minutes (mimics human thinking time)
- Max 3–5 messages before suggesting a meeting
- Detects sentiment: positive → push for meeting, negative → graceful exit
- **Human Handoff:** Notifies you when complex questions arise or user needs manual attention
- **Conversation Memory:** Maintains context across multiple exchanges
- Stops auto-replying when you send a manual message

3. Technology Stack

Layer	Technology	Reason
Desktop Shell	Electron	Cross-platform, embedded browser
Frontend	React + Vite + TypeScript	Fast, modern UI framework
Styling	Vanilla CSS + CSS Modules	Full control, no framework bloat
Browser Automation	Puppeteer + puppeteer-extra-plugin-stealth	Best stealth automation
Backend Logic	Node.js (Electron Main Process)	Same runtime as Electron
Database	SQLite (via better-sqlite3)	Local, no server needed
Task Queue	Agenda.js (MongoDB-less) or node-schedule	Job scheduling without Redis
Microsoft Auth	MSAL.js (Microsoft Auth Library)	Official OAuth2 for desktop apps

Microsoft API	@microsoft/microsoft-graph-client	Official Graph SDK
AI Integration	OpenAI SDK / Anthropic SDK	Personalization engine
Encryption	electron-store + keytar	Secure credential storage

Layer	Technology	Reason
Desktop Shell	Electron	Cross-platform, embedded browser
Frontend	React + Vite + TypeScript	Fast, modern UI framework
Styling	Vanilla CSS + CSS Modules	Full control, no framework bloat
Browser Automation	Puppeteer + puppeteer-extra-plugin-stealth	Best stealth automation
Backend Logic	Node.js (Electron Main Process)	Same runtime as Electron
Database	SQLite (via better-sqlite3)	Local, no server needed
Task Queue	Agenda.js (MongoDB-less) or node-schedule	Job scheduling without Redis
Microsoft Auth	MSAL.js (Microsoft Auth Library)	Official OAuth2 for desktop apps
Microsoft API	@microsoft/microsoft-graph-client	Official Graph SDK
AI Integration	Ollama REST API (self-hosted at `35.175.238.52`)	Primary AI — Mixtral 8x7B + Mistral 7B
Encryption	electron-store + keytar	Secure credential storage

4. Campaign Workflow (End-to-End)

A typical outreach campaign flow:

```plaintext
1. Import Leads (LinkedIn Search / Sales Nav Search / CSV)
```

```plaintext
1. Import Leads (LinkedIn Search / Sales Nav Search / CSV)
        ↓
2. Enrich Profiles (Visit + scrape data naturally over days)
        ↓
3. AI Generates Personalized Content (notes, emails, sequences)
        ↓
4. Send Connection Requests (with notes where possible)
        ↓
5. Simultaneously: Send Personalized Email (O365)
        ↓
6. Monitor: Did they accept? (Check daily)
        ↓
  ┌─ YES: Send welcome DM → Chatbot takes over → Guide to meeting
  └─ NO (3 days): Send follow-up email → Retry connection
        ↓
7. Chatbot Conversation → Detect buying intent → Book MS Teams meeting
        ↓
8. Meeting Booked → Send confirmation → Campaign goal achieved ✅
```

5. Dashboard Features

Screen	Features
Campaign Builder	Create multi-step drip campaigns with conditions/delays
Lead Pipeline	Kanban board: New → Contacted → Connected → Engaged → Meeting Booked
Analytics	Acceptance rates, reply rates, email open rates, meeting conversion
Content Calendar	Visual scheduler for posts with drag-and-drop
Inbox	Unified view of LinkedIn DMs + email conversations
Settings	Account config, daily limits, working hours, AI provider keys
Activity Log	Full audit trail of all automated actions

6. Project Structure

plaintext

plaintext

```
virtulinkedAI/
├── package.json
├── electron.vite.config.ts
├── src/
│   ├── main/                    # Electron main process
│   │   ├── index.ts             # App entry point
│   │   ├── browser/             # Anti-detect browser engine
│   │   │   ├── engine.ts        # Puppeteer stealth setup
│   │   │   ├── humanizer.ts     # Human-like behaviors
│   │   │   ├── fingerprint.ts   # Browser fingerprint management
│   │   │   └── session.ts       # Session management
│   │   ├── linkedin/            # LinkedIn interaction layer
│   │   │   ├── scraper.ts       # Profile data extraction
│   │   │   ├── connector.ts     # Connection request manager
│   │   │   ├── messenger.ts     # DM management
│   │   │   ├── navigator.ts     # Sales Navigator support
│   │   │   └── feed.ts          # Feed interaction (like, comment)
│   │   ├── microsoft/           # Microsoft 365 integration
│   │   │   ├── auth.ts          # MSAL OAuth2 flow
│   │   │   ├── email.ts         # Email sending via Graph API
│   │   │   ├── calendar.ts      # Meeting booking
│   │   │   └── tracking.ts      # Email open/click tracking
│   │   ├── ai/                  # AI personalization
│   │   │   ├── personalizer.ts  # Content generation
│   │   │   ├── chatbot.ts       # Auto-reply logic
│   │   │   ├── templates.ts     # Prompt templates
│   │   │   └── sentiment.ts     # Intent/sentiment analysis
│   │   ├── campaigns/           # Campaign orchestration
│   │   │   ├── engine.ts        # Campaign state machine
│   │   │   ├── scheduler.ts     # Action scheduling
│   │   │   ├── conditions.ts    # Conditional logic
│   │   │   └── sequences.ts     # Email/DM sequences
│   │   ├── content/             # Content scheduling
│   │   │   ├── scheduler.ts     # Post scheduling
│   │   │   ├── engagement.ts    # Auto-engagement logic
│   │   │   └── analytics.ts     # Post performance tracking
│   │   └── storage/             # Data layer
│   │       ├── database.ts      # SQLite setup
│   │       ├── models.ts        # Data models
│   │       └── migrations.ts    # Schema migrations
│   ├── renderer/                # Electron renderer (React UI)
│   │   ├── index.html
│   │   ├── App.tsx
│   │   ├── pages/
│   │   │   ├── Dashboard.tsx
```

```
│   │   │   ├── Dashboard.tsx
│   │   │   ├── Campaigns.tsx
│   │   │   ├── Leads.tsx
│   │   │   ├── ContentCalendar.tsx
│   │   │   ├── Inbox.tsx
│   │   │   ├── Analytics.tsx
│   │   │   └── Settings.tsx
│   │   ├── components/
│   │   └── styles/
│   └── preload/                # Electron preload scripts
│       └── index.ts
└── resources/                  # App resources
```

7. Safety & Anti-Detection Summary

> CAUTION
> **Critical Safety Measures — Non-Negotiable in Implementation:**

1. **No LinkedIn API usage** — all interactions via browser simulation
2. **No code injection** — pure DOM interaction like a real user
3. **In-page navigation** — never direct URL access after initial login
4. **Gaussian randomized delays** — never fixed timing patterns
5. **Account warm-up** — 2-week ramp-up for new accounts
6. **Working hours only** — no 3AM activity
7. **Daily limit randomization** — ±15% variance every day
8. **Session consistency** — same browser fingerprint, same IP
9. **Natural scrolling/reading** — profiles are "read" before action
10. **Engagement quality** — AI ensures high-quality, unique interactions

8. Phased Delivery Plan

Phase	Scope	Estimated Timeline
Phase 1	Electron app + Anti-detect browser + LinkedIn login/session	Week 1–2
Phase 2	Profile scraping + Connection requests with AI notes	Week 2–3
Phase 3	O365 Email integration + Email sequences	Week 3–4
Phase 4	Auto-reply chatbot + Calendar booking	Week 4–5

Phase 4	Auto-reply chatbot + Calendar booking	Week 4–5
Phase 5	Content scheduler + Engagement automation	Week 5–6
Phase 6	Dashboard UI + Analytics + Polish	Week 6–7
Phase 7	Sales Navigator support + Advanced campaigns	Week 7–8

9. Prerequisites (What You'll Need)

Requirement	Purpose
Azure AD App Registration	O365 email & calendar access
OpenAI or Anthropic API Key	AI personalization
LinkedIn Account	Normal and/or Sales Navigator
Node.js 18+	Runtime
macOS/Windows/Linux	Desktop platform

Requirement	Purpose
Azure AD App Registration	O365 email & calendar access
Ollama Server	Self-hosted at `35.175.238.52` (already running ✅)
LinkedIn Account	Normal and/or Sales Navigator
Node.js 18+	Runtime
macOS/Windows/Linux	Desktop platform

User Review Required

> IMPORTANT
> **Please review and confirm the following before we begin execution:**

1. **AI Provider Preference:** Do you prefer OpenAI (GPT-4) or Anthropic (Claude) for AI

1. **AI Provider Preference:** Do you prefer OpenAI (GPT-4) or Anthropic (Claude) for AI personalization? Or should we support both?
2. **Azure AD Setup:** Do you already have an Azure AD app registered for Graph API, or should I include setup instructions?
3. **Priority Order:** Should I follow the phased plan above, or would you like to prioritize specific features?
4. **Email Tracking:** Do you want email open/click tracking via tracking pixels, or is that not needed?
5. **Multi-Account:** Do you need to manage multiple LinkedIn accounts simultaneously?
6. **Content Types:** For the content scheduler, which post types are most important to you? (text, images, carousels, polls, articles, video?)
7. **Language:** Will your outreach be primarily in English, or do we need multi-language support?

1. ~~**AI Provider Preference:**~~ ✅ **Resolved** — Using self-hosted Ollama (Mixtral 8x7B primary, Mistral 7B fallback) at `35.175.238.52`
2. **Azure AD Setup:** Do you already have an Azure AD app registered for Graph API, or should I include setup instructions?
3. **Priority Order:** Should I follow the phased plan above, or would you like to prioritize specific features?
4. **Email Tracking:** Do you want email open/click tracking via tracking pixels, or is that not needed?
5. **Multi-Account:** Do you need to manage multiple LinkedIn accounts simultaneously?
6. **Content Types:** For the content scheduler, which post types are most important to you? (text, images, carousels, polls, articles, video?)
7. **Language:** Will your outreach be primarily in English, or do we need multi-language support?

Verification Plan

Automated Testing

- Unit tests for each module using **Vitest** (bundled with electron-vite)
- Integration tests for Microsoft Graph API calls using mock server
- E2E browser tests for LinkedIn interactions in a sandbox environment
- Run: `npm test` (after project setup)

Manual Verification

- **Anti-Detection Test:** Run the browser through bot.sannysoft.com and browserleaks.com to verify fingerprint matches a real browser
- **LinkedIn Safety:** Start with a test LinkedIn account, run at minimum limits for 1 week, monitor for any warnings

- **Anti-Detection Test:** Run the browser through bot.sannysoft.com and browserleaks.com to verify fingerprint matches a real browser
- **LinkedIn Safety:** Start with a test LinkedIn account, run at minimum limits for 1 week, monitor for any warnings
- **Email Flow:** Send test emails via O365, verify delivery, open tracking
- **Chatbot:** Simulate conversation flows, verify natural response timing and meeting booking
- **UI Review:** You manually review each dashboard screen for usability